AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-25 | Entry window 25 October 2026 (full day) | 3 records screened in scope

3	3	0	0	1
RECORDS IN SCOPE (52 EXCLUDED)	OF 10 SUBTOPIC BANDS POPULATED	EFFECT ESTIMATES WITH A CI	TIER-A RECORDS	RECORDS CARRIED ON METADATA ALONE

Signal of the day

A Sunday floor issue: three records, and the most watch-relevant point is that 3-D printing emits almost entirely below the optical cut of consumer PM sensors.

Recycled plastic-bag HDPE extruded to filament and then 3-D printed (NIOSH chamber work): extrusion emits mostly 1.2–5.2 nm particles, printing mostly 5.6–560 nm, with number yields per gram **2–3 orders of magnitude** higher during printing; formaldehyde 9.9–16.1 $\mu g\,g^{-1}$ printed. **Where it is weak:** yields are per mass processed in an enclosure, not breathing-zone concentrations, and no uncertainty is attached to the contrast. **Why it matters here:** a PMS-class optical node registers almost none of this number emission; maker-space and indoor-occupational monitoring needs a condensation or diffusion-charging counter.

What changed today

- **Thin day, heavy overlap.** Ten of the 13 records that passed the topic screen had already been issued on 23, 24 or 26 October (Europe PMC re-surfacing PubMed entries one to two days late) and are logged as duplicates.
- **Single-monitor ceiling.** Cubatão schoolchildren ($n = 77$): HRV and spirometry associations are for SO_2, O_3 and NO_2 assigned from one regulatory station; no PM association is reported.
- **Street-scale morphology thresholds** for $PM_{2.5}/PM_{10}$ (Atmos. Pollut. Res.) arrived as a metadata-only Crossref deposit.

Window continuity

The 22 October issue closed with `last_entry_date` at 22 October. The scheduled runs for 23–29 October did not complete (a 30 October 00:21 UTC run harvested 23–28 October but stopped before screening). This issue was built on the **30 October 14:20 UTC (09:20 CDT) recovery run**, which backfills 23–29 October day by day; it collects **25 October in full** (Sunday), continuous with the 24 October issue.

Corpus analytics

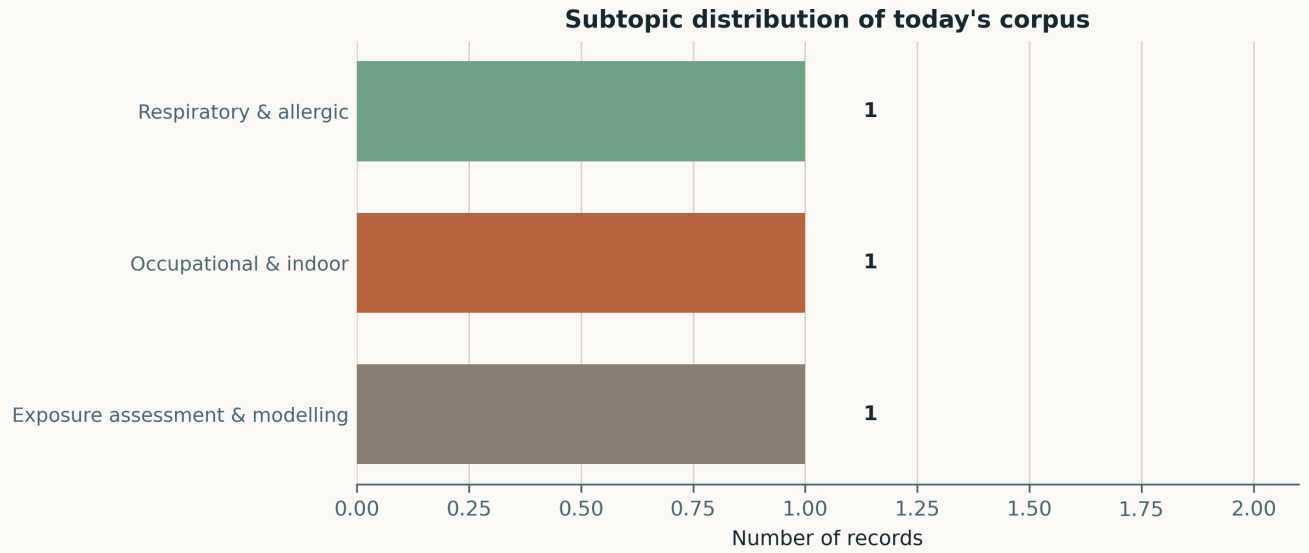


Fig. Subtopic assignment of the 3 in-scope records: respiratory (Cubatão schoolchildren), occupational & indoor (3-D printing emissions) and exposure assessment (street-scale morphology, metadata-only).

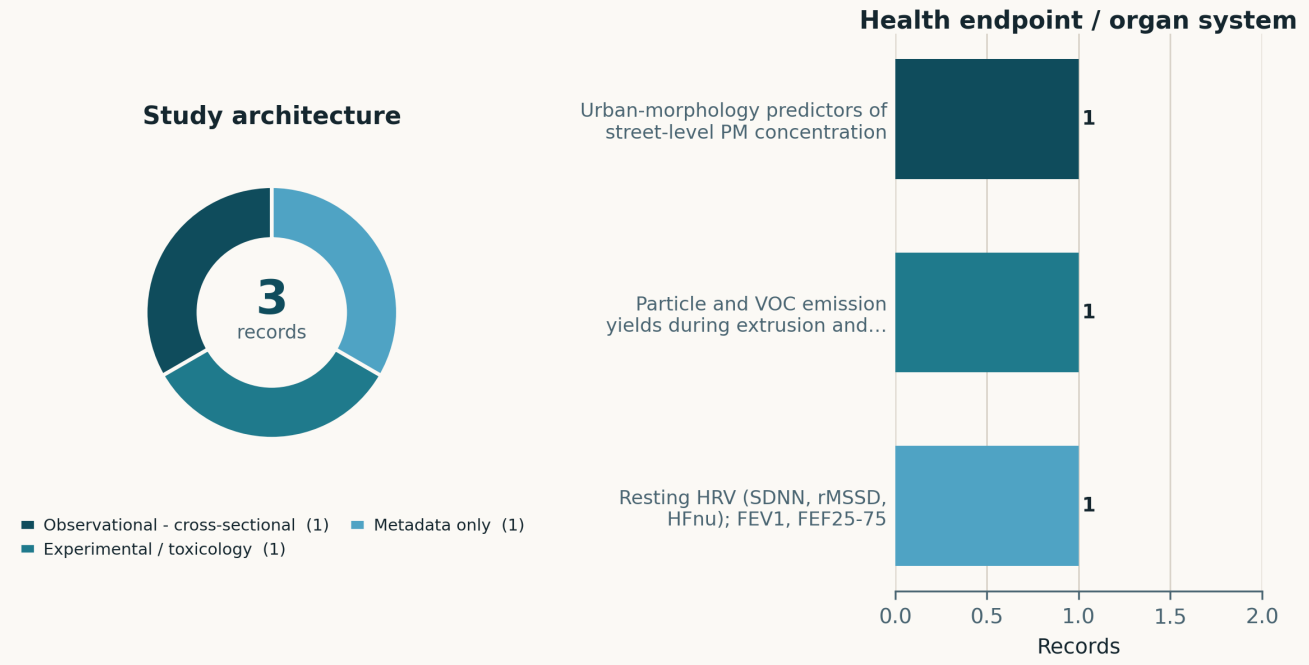


Fig. Left — architecture: one school-based cross-sectional study, one chamber emission characterisation, one metadata-only record (1 of 3). Right — endpoints: HRV/spirometry, emission yields, and street-level concentration.

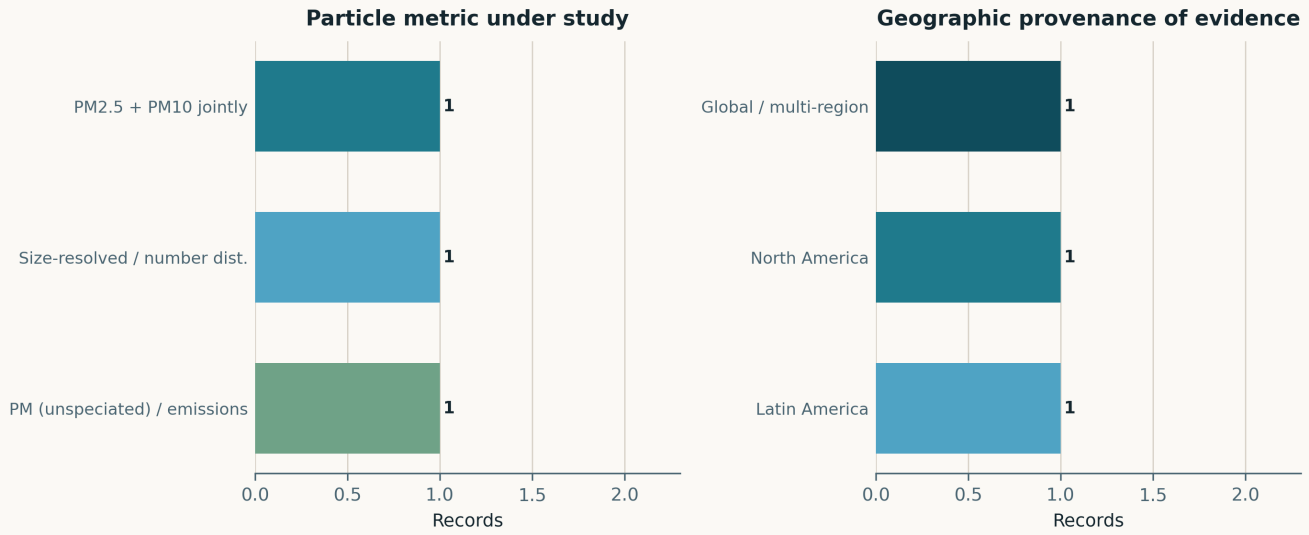


Fig. Left — particle metric: regulatory-monitor pollutants, nanoscale number emissions, and $PM_{2.5}$+PM_{10}. Right — geography: Brazil, United States (laboratory), and one not stated.

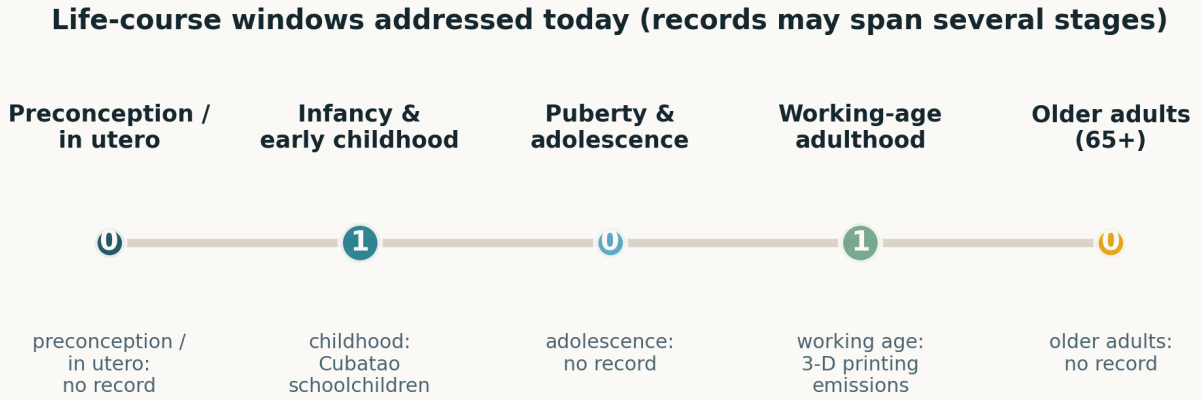


Fig. Life-course windows: one childhood record (Cubatão) and one working-age occupational record; no in-utero, adolescent or older-adult record today.

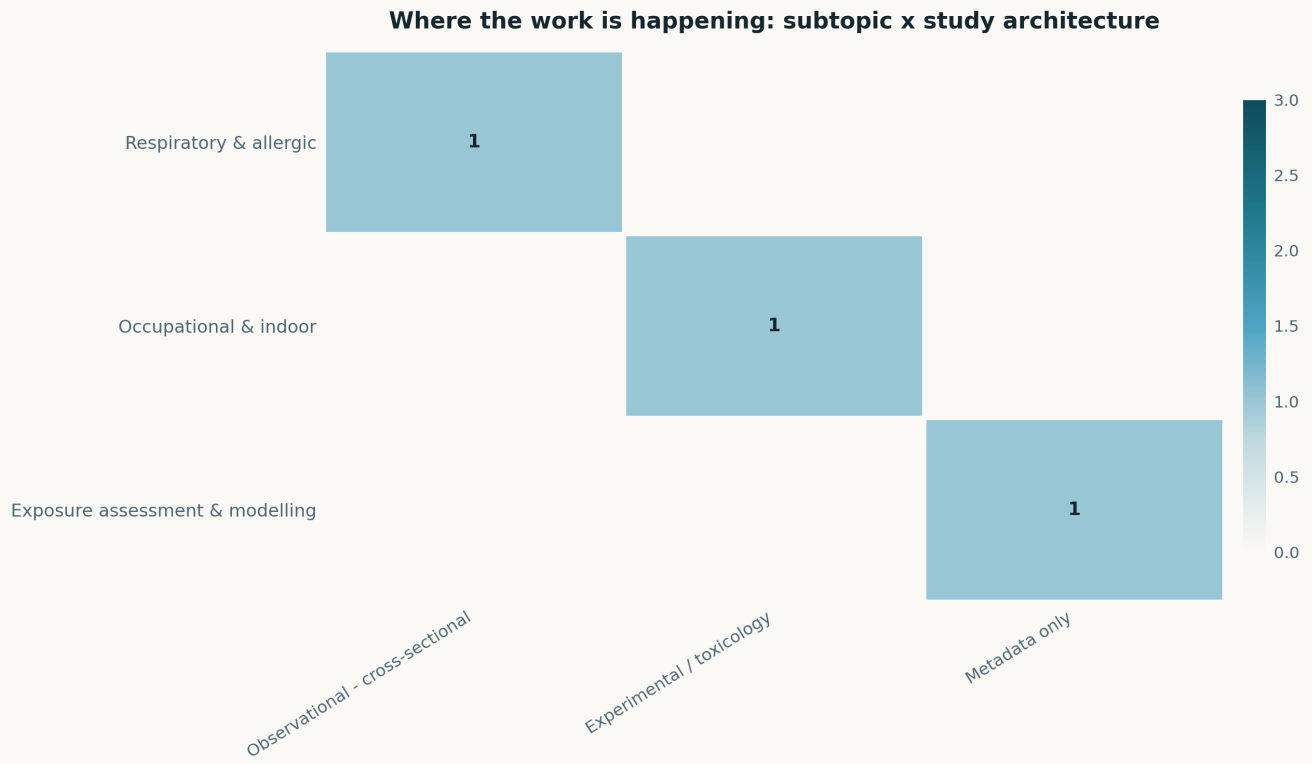


Fig. Subtopic × study architecture. Three singleton cells; no cohort, trial or effect estimate with a confidence interval today, so no forest plot is drawn.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Respiratory & allergic — 1 record

Oliveira air pollution HRV lung function schoolchildren Cubatao 2026 Epidemiol Serv Saude

Effect of air pollution on heart rate variability and pulmonary function in schoolchildren: a cross-sectional study, Cubatao, Brazil, 2017.

Convenience sample of 77 schoolchildren in the Cubatao industrial zone, with pollutant levels from the state agency's monitors. Associations reported in the abstract are for gases: SO_2 and HFnu -0.68 (-1.27 to -0.11) daily, -1.29 (-2.21 to -0.38) over three days; O_3 and HFnu -0.32 (-0.62 to -0.02); NO_2 with 0.23% lower FEV1 and 0.30% lower FEF25-75 (no CI given). No PM association is reported in the abstract. Limitation: 77 children, one fixed-site monitor and a single visit leave essentially no exposure contrast across subjects - the estimates rest on day-to-day variation within a cross-sectional design, with multiple pollutants and lags tested. Relevance: low - illustrates the exposure-assignment ceiling of a single regulatory monitor.

doi: 10.1590/s2237-96222026v35e20250368.en

Exposure assessment & modelling — 1 record

Wang urban morphology street-scale $PM_{2.5}$ PM_{10} change points 2026 Atmos Pollut Res

Non-linear associations, model-supported change points, and seasonal variations of urban morphology with street-scale $PM_{2.5}$ and PM_{10} concentrations

METADATA ONLY. Metadata-only record: Crossref deposit with no abstract on the day of entry, and no abstract recoverable from Europe PMC or Semantic Scholar. The title reports non-linear associations and model-supported change points between urban morphology indicators and street-scale $PM_{2.5}$ and PM_{10}, with seasonal variation. Design, city, sample size, morphology metrics and effect sizes cannot be stated from the record and are not inferred here. Limitation: unassessable until the abstract or full text is available; threshold-type 'change points' in morphology-PM regressions are prone to overfitting when sites are few. Relevance: medium - street-scale morphology thresholds bear directly on how a dense sensor network should be sited across canyon geometries.

doi: 10.1016/j.apr.2026.103242

Occupational & indoor 1 record

Stefaniak recycled plastic bag HDPE filament 3-D printing emissions 2026 J Chem Health Saf

Recycling Plastic Bags into Filament: Consideration of Emissions during Melting, Extrusion, and 3-D Printing.

Waste HDPE plastic bags from South Africa and the US were blended with virgin HDPE, extruded into filament and 3-D printed, with particle- and gas-phase emissions measured through each stage. Filament extrusion emitted mostly 1.2-5.2 nm particles; printing emitted mostly 5.6-560 nm particles, with particle yields (no./g processed) 2-3 orders of magnitude higher than extrusion. Printing also released acetone (1.4-39.4 ug/g printed) and formaldehyde (9.9-16.1 ug/g). Mechanical properties matched virgin HDPE. Limitation: yields are per mass processed in a controlled enclosure, not personal breathing-zone concentrations, and no uncertainty is attached to the order-of-magnitude contrast; feedstock contamination of bag plastic was not characterised. Relevance: low-medium - nearly all the number emission sits below the ~300 nm optical cut of consumer PM sensors, so maker-space monitoring needs a CPC or diffusion charger, not a PMS-class node.

doi: 10.1021/acs.chas.5c00138

Corpus register

First author	Focus	Journal	Design	Metric	Region
Respiratory & allergic					
Oliveira air pollution HRV lung function schoolchildren Cubatao	Effect of air pollution on heart rate variability and pulmonary function in schoolchildren: a cross-sectional study, Cubatao, Brazil, 2017.	*Epidemiol Serv Saude*	School-based cross-sectional	Regulatory monitor concentrations (CETESB), daily and 2-3 day averages	Brazil (Cubatao, Sao Paulo State)
Occupational & indoor					
Stefaniak recycled plastic bag HDPE filament 3-D printing emissions	Recycling Plastic Bags into Filament: Consideration of Emissions during Melting, Extrusion, and 3-D Printing.	*J Chem Health Saf*	Chamber emission characterisation	Particle number size distribution (1.2-560 nm), particle yields per g plastic; acetone and formaldehyde	United States (laboratory; South African and US HDPE bags)
Exposure assessment & modelling					
Wang urban morphology street-scale $PM_{2.5}$ PM_{10} change points	Non-linear associations, model-supported change points, and seasonal variations of urban morphology with street-scale $PM_{2.5}$ and PM_{10} concentrations	*Atmos Pollut Res*	Metadata only (no abstract)	Street-scale $PM_{2.5}$ and PM_{10}	Not stated (metadata only)

Provenance and method

Entry window **25 October 2026**, single day (Sunday). Harvester legs (run 30 October 00:21 UTC): **PubMed** health **12**, sensing **1**; **Europe PMC 68**; **Crossref by ISSN 7**; **arXiv** 150 kB, no in-window entry; **OpenAlex** HTTP 429. PubMed connector (EDAT 23–28 Oct, one backfill query bucketed by entry date) appended 1. **55 screened**. **3 admitted**, **52 rejected** with logged reasons, 10 of them cross-day duplicates of the 23, 24 and 26 October issues. **DOI gate** 0 fail / 0 warn. **0 effect estimates with a CI** — no forest plot. **1 carried on metadata alone**. Consensus sweep and trial watch were run once for the 23–29 October backfill and are reported in the 29 October issue. Abstracts remain the copyright of their publishers.